First Level 1 Product Results of the Greenhouse Gas Monitoring Instrument on the GaoFen-5 Satellite

Hailiang Shi, Zhiwei Li, Hanhan Ye, Haiyan Luo, Wei Xiong, and Xianhua Wang

***Abstract*— The Greenhouse Gas Monitoring Instrument (GMI) is a short-wavelength infrared (SWIR) hyperspectral-resolution spectrometer onboard the Chinese satellite GaoFen-5 that uses a spatial heterodyne spectroscopy (SHS) interferometer to acquire interferograms. The GMI was designed to measure and study the source and sink processes of carbon dioxide and methane in the troposphere where the greenhouse effect occurs. In this study, the processing and geometric correction algorithms of the GMI Level 1 product (radiance spectrum) are introduced. The spectral quality and greenhouse gas (GHG) inversion ability of the Level 1 products are analyzed, and the results illustrate that the specifications meet the mission's requirements. An initial evaluation of the resolution, signal-to-noise ratio (SNR), and stability of the radiance spectrum reveals that the overall function and performance are within the design objectives. A comparison between our Level 1 products and the theoretical spectrum shows that the root mean square (rms) of the residual is approximately 0.8%, and the Level 1 products of the GMI captured within five months after observations have good spectral stability characteristics (less than 0.005 cm^{-1} for Band 1, 0.003 cm^{-1} for Band 2, 0.002 cm^{-1} for Band 3, and 0.004 cm^{-1} for Band 4). These results demonstrate that the GMI payload and the processing algorithm all work well and reliably. Furthermore, based on the Level 1 products, a GHG retrieval experiment is carried out, and the results are compared with data from Total Column Carbon Observing Network (TCCON) stations. The initial comparison of the XCO_2 results yields a value of 0.869 for R^2 (goodness of fit), 0.51 ppm for bias (mean of absolute error), and 0.53 ppm for σ (standard deviation of error). Similarly, the XCH_4 comparison yields values of 0.841 for R^2, 4.64 ppb for bias, and 4.66 ppb for σ.**

***Index Terms*— GaoFen-5 (GF-5) satellite, greenhouse gas (GHG), Greenhouse Gas Monitoring Instrument (GMI), spatial heterodyne spectrometer, spectrum.**

NOMENCLATURE

GMI	Greenhouse Gas Monitoring Instrument.
SHS	Spatial heterodyne spectroscopy.
GF-5	GAOFEN-5.
CHEOS	China High-Resolution Earth Observation System.
AIOFM	Anhui Institute of Optical Fine Mechanics.
CAS	Chinese Academy of Sciences.
GHGs	Greenhouse gases.
TCCON	Total Column Carbon Observing Network.
FFT	Fast Fourier transform.
ZPD	Zero-path difference.
DC	Dark current.
FWHM	Full-width at half-maximum.
GCPs	Ground control points.
ILS	Instrumental line shape.
rms	Root mean square.
OEM	Optimal estimation method.

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I. INTRODUCTION

THE GF-5 satellite is the latest satellite to achieve hyperspectral resolution observations in the CHEOS. It has global hyperspectral observation capability and is used to obtain hyperspectral-resolution remote sensing data from ultraviolet to long-wave infrared bands [1]. It was successfully launched from the Taiyuan Satellite Launch Center in May 2018. The satellite carries six payloads. Among them, the GMI payload was developed by the AIOFM, CAS, Hefei, China [2]. The GMI is mainly used to obtain global column concentration data on the GHGs carbon dioxide (CO_2) and methane (CH_4).

Due to the small variation in GHG concentrations spatially and temporally, the spectral resolution and signal-to-noise ratio (SNR) of the detection instrument are required to be high. To meet the above requirements, the GMI payload on the GF-5 satellite adopts a novel form of spectroscopy, namely, SHS, to obtain the atmospheric absorption spectrum. The GMI is also the first GHG payload based on SHS and operates in a solar synchronous orbit with a nominal height of 705 km. Because the mechanism of the GMI is different from TANSO-Fourier transform spectroscopy (FTS)/GOSAT (Michelson interferometer) [3] and OCO-2 (Grating) [4], the data processing algorithms are quite different.

The main objective of this article is to introduce the processing algorithm for the Level 1 product (radiance spectrum) developed for the GMI and to discuss the first results of the hyperspectral-resolution spectrum from the GMI. In Section II, the GMI, the corresponding measurement characteristics, and the data products are presented. Section III describes the Level 1 processing algorithms, including the data screening (DS) algorithm, the interferogram processing (IP) algo-

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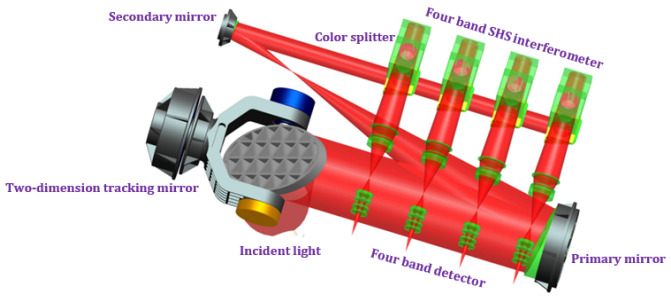


Fig. 1. Optical configurations of the GMI.

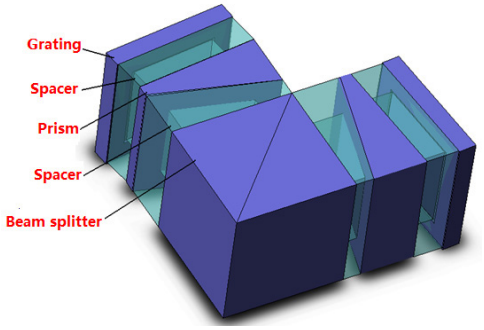


Fig. 2. SHS interferometer.

rithm, and the spectrum recovery (SR) algorithm, in detail. Section IV describes the geometric correction of the GMI Level 1 product based on the special surface of the coastline. In Section V, the first results of the Level 1 products are introduced in detail, including the onboard spectral resolution and SNR specifications, and a comparison with the theoretical spectrum and an analysis of the spectral stability are presented. Then, preliminary retrieval results for XCO_2 and XCH_4 retrieved from GMI measurements are compared with the TCCON products.

II. Overview of the GMI

A. Instrument

The GMI is an SHS interferometer used to measure the atmospheric absorption spectrum. Compared with a conventional Michelson interferometer, the SHS interferometer replaces the moving mirrors with diffraction gratings, resulting in extremely high spectral resolution within a narrow band [5], [6]. The GMI optical configuration is shown in Fig. 1. The incident light is reflected by a 2-D tracking mirror and enters the Gregory reflector telescope. Then, the beam diameter is compressed and enters the color splitter. It is divided into four bands, and each band of the incident light is modulated by an SHS interferometer composed of a beam splitter, prism, grating, and spacer (see Fig. 2). The interferogram is imaged by the imaging lens with a charge-coupled device (CCD) detector (760-nm band), two InGaAs detectors (1575- and 1640-nm bands), and a movable core transformer (MCT) detector (2050-nm band). The main optical system of the GMI is divided into five components according to the function: telescope system, color separation module, collimation relay system, spatial heterodyne interferometer, and imaging system.

To improve the data accuracy, the GMI is equipped with an onboard calibration unit, as shown in Fig. 3, using high-stability sunlight as the calibration light source. The coordinate system is defined with X as the direction of satellite flight, Z as the direction of Earth's observation, and Y as the right-hand rule direction. When the GMI is going to carry out a calibration, the 2-D tracking mirror is adjusted from the nadir/sun-glint observation mode to the calibration mode. The calibration is performed over the Antarctic. First, the spectral calibration is done (the sunlight passes through the atmosphere and the altitude is 10–100 km), and then the radiometric

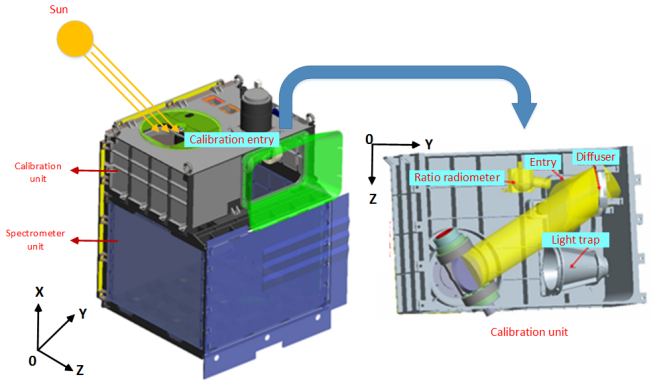


Fig. 3. Calibration unit of the GMI.

calibration is carried out (the sunlight does not pass through the atmosphere and the altitude is greater than 100 km). To meet the requirements of on-orbit spectral and radiometric calibration, the calibration unit has a designed diffuser, light trap, ratio radiometer, and other components, which are used to ensure the uniformity of the sunlight, monitor the decay of the diffuser, and obtain the DC of the instrument.

B. Specifications

The GMI includes the nadir and sun glint models, similar to the GOSAT satellite. The nadir model takes nominal observations over dayside land, using sampled points along the cross-track direction as footprints. However, the sun-glint model mainly solves the problem in which the reflected signal from the ocean is weak and can track the sea flare automatically in real time according to the parameters of the sun, satellite position, and so on. As shown in Fig. 4, T0 is the start of solar illumination of the satellite, T1 to T2 is the spectral calibration time, T2 to T3 is the radiation calibration time, and T5 to T6 is the nadir or sun-glint observation mode time.

The GMI records the solar light reflected from the Earth's surface from the near-infrared (NIR) wavelengths to the shortwave infrared (SWIR) region and has one NIR band (centered at 0.76 μm, known as GMI Band 1) and three SWIR bands (centered at 1.58, 1.65, and 2.0 μm, known as GMI Bands 2, 3, and 4, respectively). The NIR band contains an oxygen (O_2) band (Band 1) that provides information on the surface

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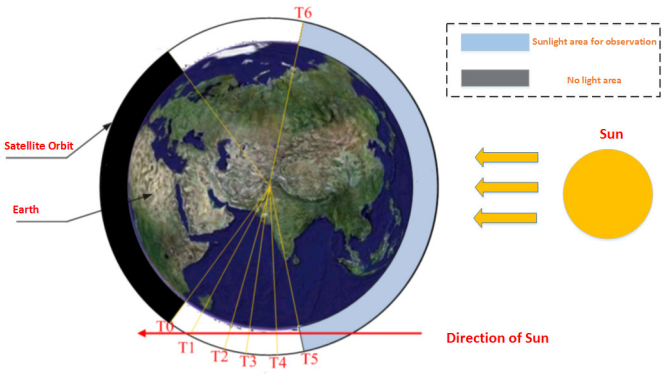


Fig. 4. Onboard observation mode of the GMI.

pressure, clouds, and aerosols. In addition, Band 2 and Band 3 provide important data regarding the weak absorption bands of CO_2 and CH_4 and provide information on the CO_2 and CH_4 columns with high near-surface sensitivity. In contrast, Band 4 measures the strong absorption of CO_2 and accordingly provides additional CO_2 information. The main specifications of GMI are shown in Table I.

C. Products

The original data obtained by the GMI are in the form of an interferogram (Level 0), which is collected by an array detector. The radiance spectrum (Level 1) of the target gas is obtained through the processing of error correction, Fourier transform, spectral calibration, and radiation calibration of the satellite downloaded detection data. The atmospheric inversion of spectral data based on the optimization iteration algorithm can yield the global column concentration distribution data of CO_2 and CH_4 (Level 2), draw a global concentration map, and estimate the global carbon flux (emission and source) (Level 3) and 3-D distribution of CO_2 and CH_4 after analyses using the atmospheric transport model.

The Level 0 data format is Raw. The Level 1, Level 2, and Level 3 data formats are HDF5 format. The data products include not only scientific observation data but also satellite platform parameters, payload state parameters, calibration coefficients, and other auxiliary data. This information is convenient for users to process the production of related scientific data products. Fig. 5 shows an overview of the overall data processing flow with the data level.

III. Level 1 Processing Algorithms

Although the background concentrations of GHGs are not low, the fluctuation is small. To achieve detection accuracies of 1% (CO_2) and 2% (CH_4), high-sensitivity instrumentation and high-precision data processing algorithms are needed. In terms of the characteristics of the instrument, algorithms should be developed according to the data error mechanism. GOSAT and OCO-2 adopt a Michelson interferometer and grating technology, respectively, while the GMI adopts the new technique of SHS, which results in a difference in data processing methods. Spatial heterodyne spectrometer's response error of array detector affects the interference fringe directly, and each interference point corresponds to the result of both pixel's response and light path response error. Grating flaws, surface-shape error, and distortion of imaging system can all probably bring in more complex phase error. Generally, spatial heterodyne spectrometer realizes hyperspectral resolution in a rather narrow band, which results in complex spectral processing flow. The GMI acquisition data (not spectrum data, which can be directly used for inversion) acquired using the array detector and interferometer are easily affected by bad pixels, nonuniformity, nonlinearity, and other factors. Here, a complete set of data processing algorithms are presented which have been implemented in the ground data processing system of the GF-5 satellite [7]–[10], but there is no systematic research on the algorithm of onboard data processing. According to the characteristics of GMI observation data, a complete set of data processing algorithms have been developed and put into the ground processing system of the GF-5 satellite.

The Level 1 data processing process is mainly divided into three modules: DS, IP, and SR, as shown in Fig. 6. First, the DS module checks Level 0 data (interferogram) through a series of judgments and eliminates the abnormal data caused by various issues. In the IP module, various interferogram errors are corrected. Finally, the SR module carries out FFT to obtain the radiance spectrum, and Level 1 data are produced, which can be used for atmospheric GHG inversion.

A. Data Screening

There are approximately 560 observation points per orbit in the nadir five-point mode, so a large amount of interferogram data are collected every day. Only the data with high-quality would be processed during GMI Level 1 product processing. High-quality data are sifted from low-quality data by DS, which mainly aims to eliminate the Level 0 product, which cannot be used to carry out SR. The criteria of DS are often conditions based. DS should be performed using individual quality criteria or a combination of criteria. Multiple data quality flags, primarily relevant to satellite and onboard instrument operations, are defined and stored in the GMI Level 1 products. The interferogram data product should be evaluated and filtered by referring to these data quality flags, such as, ZPD, SNR, Nosie, and Saturation. The following is a description of data filtering through the use of standards designed to define data quality. Thus, the GMI interferogram data should be judged to be of low quality and subject to being filtered out if they cannot satisfy at least one of the following criteria.

1) ZPD Deviation: The ZPD position in the interferogram is detected and compared with the normal value. When the deviation is less than a certain threshold (two sampling points), the ZPD position is correct.

2) Low SNR: When the sampled footprints have low reflectivity, the obtained interferogram has a low SNR and cannot meet the requirements of SR. By extracting the DC component of the interference fringes, it can be judged that the data SNR are low when the digital number (DN) values of Band 1 to Band 4 are less than 1000, 1500, 1500, and 2000, respectively.

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TABLE I
SPECIFICATIONS OF THE GMI

Specifications	Values			
	Band 1	Band 2	Band 3	Band 4
Detected gas	O_2	CO_2	CH_4	CO_2
Band range (μm)	0.759-0.769	1.568-1.583	1.642-1.658	2.043-2.058
Spectral resolution (cm^{-1})	0.6	0.27		
SNR (albedo=0.3; sun elevation=30°)	300		250	
Radiometric calibration	absolute accuracy: 5%, relative accuracy: 2%			
FOV	14.6 mrad (10.3 km @ 705 km)			
Scan	Cross track (± 35 deg), Along track (± 20 deg)			
Observation modes	Nadir: 1, 5, 7, 9 points (default mode is 5 points); Sun glint; Calibration			
Number of detector pixels	1024×1024	512×640	512×640	256×320
Zero optical path difference pixel	512	235	235	75
Sampling form	Symmetric	Asymmetric	Asymmetric	Asymmetric

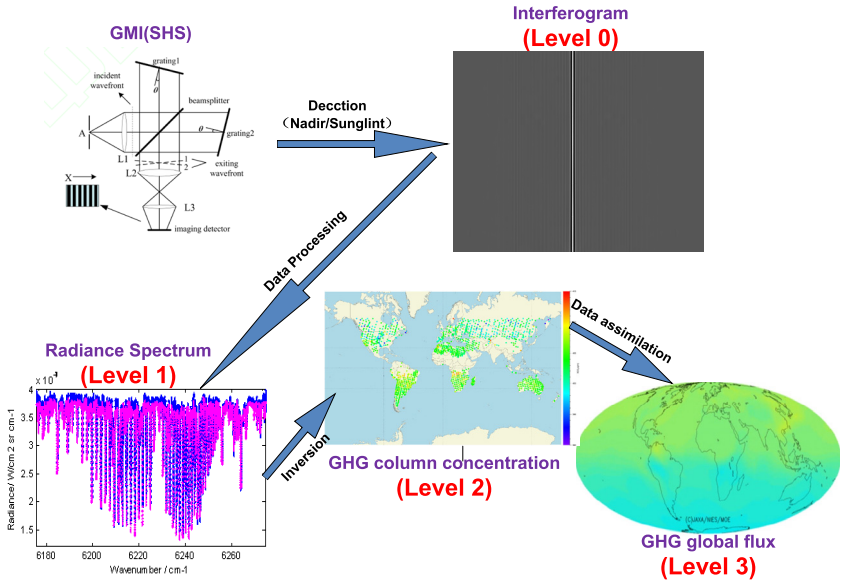


Fig. 5. Data processing flowchart and data product levels.

3) Saturation Detection: When the observation point is located in an area with high reflectivity or cloud coverage, the interferogram is saturated. Similarly, the DN value of the DC component of the interference fringes is used for judgment. If the value is greater than 85% of the dynamic range of the band, the data are considered to be saturated.

4) Spike Noise: Due to the influence of the space environment, the interferogram may produce spike noise. If the noise point is near the zero optical path difference, the interferogram is judged to be invalid. In addition, if the noise is not an independent point but a continuous noise band, the data are also considered abnormal.

5) Correlation Coefficients: The interferogram removes the DC component and retains effective modulation information I_0. At the same time, the normal interferograms measured in the laboratory (or on orbit) are processed equally to obtain I_N. The correlation coefficient $r(I_0, I_N)$ is calculated using the following formula:

$$r(I_0, I_N) = \frac{\sum_{i=1}^{n} (I_{0,i} - \overline{I_0})(I_{N,i} - \overline{I_N})}{\sqrt{\sum_{i=1}^{n} (I_{0,i} - \overline{I_0})^2} \sqrt{\sum_{i=1}^{n} (I_{N,i} - \overline{I_N})^2}}. \quad (1)$$

When the correlation coefficient $r(I_0, I_N)$ is larger than the threshold value (0.9), the original Level 0 data are considered normal.

B. Interferogram Processing

The interferogram filtered by quality control still contains various random or systematic errors, which need to be corrected. In the IP module, the interferogram will be processed by DC deduction, bad pixel correction, baseline elimination, nonlinear correction, flat-field correction, phase correction, and apodization, and this process provides ideal data for the next high-precision spectral recovery. Among them, the DC deduction is processed by the observation data obtained from the light trap in the calibration process, and the nonlinear correction is processed by the detector linear curve obtained in

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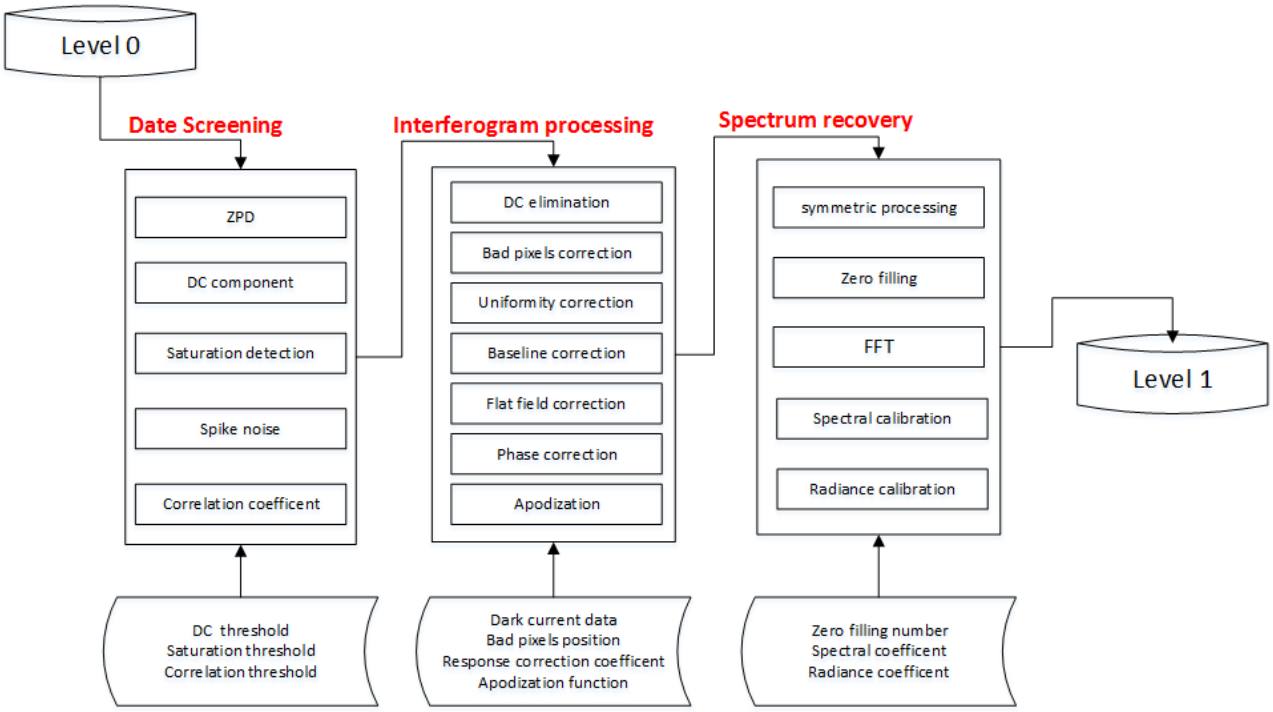


Fig. 6. Flowchart of Level 0–Level 1 data processing algorithms.

the laboratory. Other data processing algorithms are described in detail below.

1) Bad Pixel Correction: In normal circumstances, the position and number of bad pixels are basically stable during the operation of the instrument, which can be corrected in the laboratory. However, in actual operation, a large number of abnormal bad pixels (overheated pixels, dead pixels, and flicker pixels) are generated due to the decay of the detector. It will lead to uncertainty of the location of spikes pixels, and some bad pixels will appear in block distribution. Due to the uncertainty of the number and location of bad pixels, the processing algorithm is required not only to have the function of repairing but also to realize the adaptive detection of bad pixels to accurately locate the bad pixels and avoid over-recognition, especially for the detection of bad pixels in the interference fringe area.

The correction of the GMI bad pixel data can be divided into two parts. First, in the development stage of the instrument, the coordinates (x, y) of the bad pixel that has been accurately located by the detector in the laboratory are replaced by spline interpolation. Second, a new bad pixel that appears in the operation process of the instrument is located by real-time assessment of observation data.

The spatial dimension of the GMI interferogram collects all the energy in the field of view at the same time. The difference in the interferogram in the column direction can be divided into slow change caused by the efficiency and phase difference in the optical system and fast change caused by the pixel of the detector. Because each band of GMI uses a high-level space class detector, the response difference (nonuniformity) between pixels is better than 5%.Then the difference value between each row of data pixels is calculated, and the real-time location of the bad pixel position of observation data can be realized by setting the empirical threshold value for the existing data analysis. Taking GMI Band 2 data as an example, the typical observation data of payload on orbit are shown on the left of Fig. 7. The data of the 10th column and the difference between its pixel values are shown in the upper right of Fig. 7. The amplification results of typical detector bad points are shown in the bottom right of Fig. 7. The bad point data of the detector can be accurately located by the pixel value difference.

The real-time positioning of the bad points in the interferogram and the comparison results before and after correction are shown in Fig. 8. The absolute majority of bad pixel data have been effectively corrected, especially the obvious bad pixel, and the remaining few bad pixels at the threshold edge have been corrected by the average of recovery spectra of different lines of data.

2) Baseline Elimination: As mentioned above, GMI data may not only have ideal DC components and spectral components but also nonuniform high-frequency interference noise of the detector and low-frequency interference noise of the instrument response (transmittance of the optical system). All data points of GMI are sampled once and run without components during sampling, which reduces the interference that may be introduced by other components during data acquisition compared with the traditional FTS. The actual GMI instrument response to the introduction of low-frequency noise is relatively complex, and the traditional polynomial fitting method cannot remove these low-frequency components. However, because these low-frequency components are fixed, they can be removed effectively by the Butterworth low-pass filter designed for each row of data.

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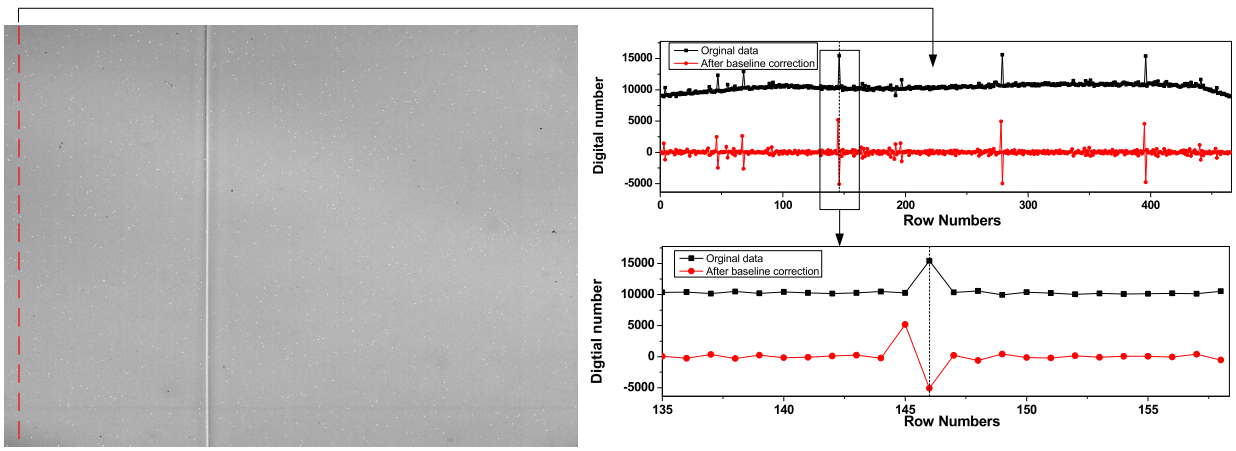


Fig. 7. Bad pixel location results of row 10 data (GMI Band 2 interferogram).

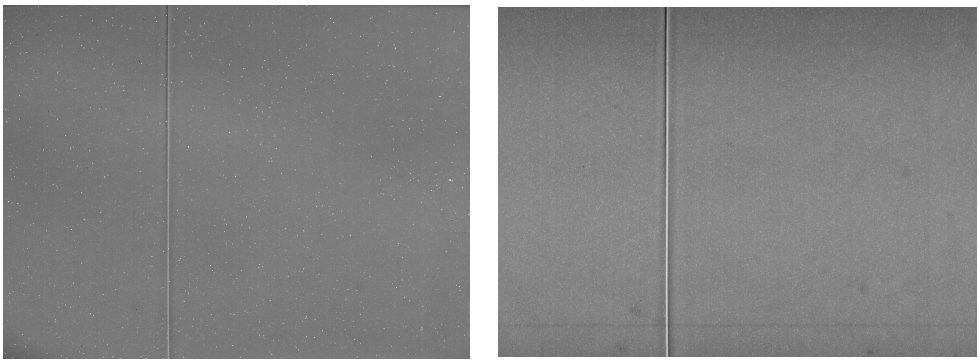


Fig. 8. Bad pixel correction result (GMI Band 2 interferogram). (Left) Before correction. (Right) After correction.

The comparison between the DC component determined by the polynomial fitting method and the DC component calculated by the Butterworth low-pass filter for one row of the GMI Band 3 interferograms are shown in Fig. 9. The filtering method can more accurately calculate and remove the low-frequency interference information in the GMI interferogram.

3) Flat-Field Correction: Compared with the Michelson interferometer, the SHS interferometer adopts the area array detector to collect the interference fringes simultaneously. This method has better stability, but at the same time, it also introduces a nonuniform response factor that the Michelson interferometer does not have. This systematic nonuniformity is mainly caused by many factors, such as the difference in the efficiency of the two arms of the interferometer, the manufacturing defects or pollution of the optical element itself (especially the grating), and the residual nonuniformity of the detector response correction, which results in an inconsistent response for each pixel in the interferogram. Therefore, the above systematic errors need to be corrected, and this process is known as flat-field correction. Traditional cameras or dispersive spectrometers use uniform light sources (such as an integrating sphere). However, for the GMI, when the incident light is uniform, the output of the instrument is the interference fringe data. Theoretically, the output response of each pixel is not the same, so the traditional flat-field method is not applicable.

To solve the problem of SHS flat-field correction, the balanced arm flat-field method has been used [11]. However, this method must be completed during the integration process of the interferometer. If the relative relationship of the optical system, structural system, or detector changes during the subsequent installation or debugging process of the instrument, the flat-field system will not match and cannot be used. Therefore, the mark points are designed to match the correction coefficient on orbit.

4) Phase Correction: The phase correction method of the GMI interferogram is the same as that of the traditional FTS, and the phase shift is calculated using the zero optical path difference in the interferogram and the data of the nearby area. The GHG observation requires the spectrometer with high enough spectral resolution capacity to meet the required detection accuracy. Therefore, the GMI (Bands 2, 3, and 4) adopts the zero optical path difference bias method when the number of pixels in the array detection spectral dimension is limited. Thus, the GMI Bands 2, 3, and 4 interference fringes are similar to the FTS zero-crossing sampling fringes.

Because the phase of the interferometer generally changes slowly, the interferogram can be corrected by the low-resolution spectral phase calculated from the short symmetry interference fringe. Short bilateral data lengths should not only meet the needs of phase resolution but also avoid the phase jitter caused by instrument noise. Taking Band 2 as an

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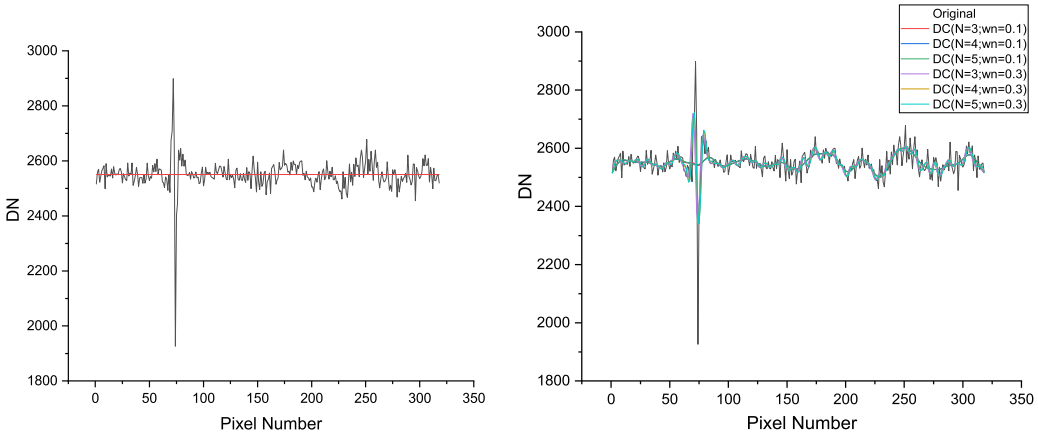


Fig. 9. Baseline elimination results (GMI Band 3 interference fringe). (Left) Polynomial fitting method. (Right) Butterworth low-pass filter method.

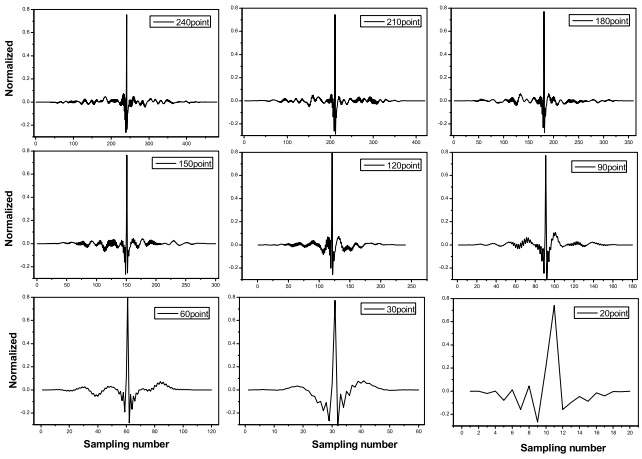


Fig. 10. Phase correction function calculated by different lengths of short symmetry interference points.

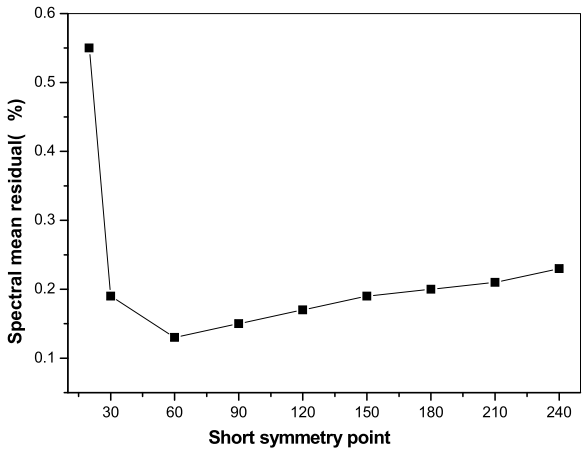


Fig. 11. Spectral mean residual based on different short symmetry points for phase correction.

example, the selection of the length of different short bilateral interference fringes is analyzed by simulation. Fig. 10 shows the phase correction function calculated by different lengths of short symmetry interference points.

The deviation between the spectrum and the theoretical spectrum after the length phase correction of different short bilateral interference data is shown in Fig. 11. According to the residual result, GMI Band 2 phase correction selects 60 short symmetrical data points.

5) Apodization: The sampling of the limited optical path difference by the interferometer results in the sidelobe oscillation of the reconstructed spectrum, so the apodization function is often used to suppress the frequency leakage in the processing of the interferogram. The application of the apodization function is to keep the spectrum of the interference spectrometer smooth and consistent with the output spectrum of other spectrometers. However, the cost of spectral smoothing is the broadening of the FWHM and the change in the energy distribution in the frequency domain, which are not directly reflected in the final data measurement accuracy [12]. Based on the criteria of the broadening of the reconstructed spectrum and the degree of sidelobe suppression, the current common apodization function does not achieve the best effect. The apodization function of the Norton beer window is shown in the following:

$$\mathrm{NB}\left(\frac{l}{U}\right)=\sum_{i=0}^{n}x_i\left(1-\left(\frac{l}{U}\right)^2\right)^i \tag{2}$$

where U is the maximum optical path difference, $l\in[-U,U]$ represents the sampling points of different optical path differences, $n=0,1,2,3\ldots$ is the number of polynomials, and the sum of the coefficients is 1, that is, $\sum_{i=0}^{n}x_i=1$. Through the iterative coefficient x_i, when the ratio of the FWHM W of the spectrum after apodization to the FWHM W_0 of the spectrum without apodization is fixed, the secondary principal maxima of the spectrum satisfy the following formula:

$$\log_{10}\left|\frac{h}{h_0}\right|=1.939-1.401\left(\frac{W}{W_0}\right)-0.597\left(\frac{W}{W_0}\right)^2. \tag{3}$$

The height of the secondary principal maxima is h_0 ($h_0=0.21723$), and the height of the secondary principal maxima is h.

As shown in Fig. 12 (left), the corresponding half height spread after the Fourier transform is 1 to 2 times larger

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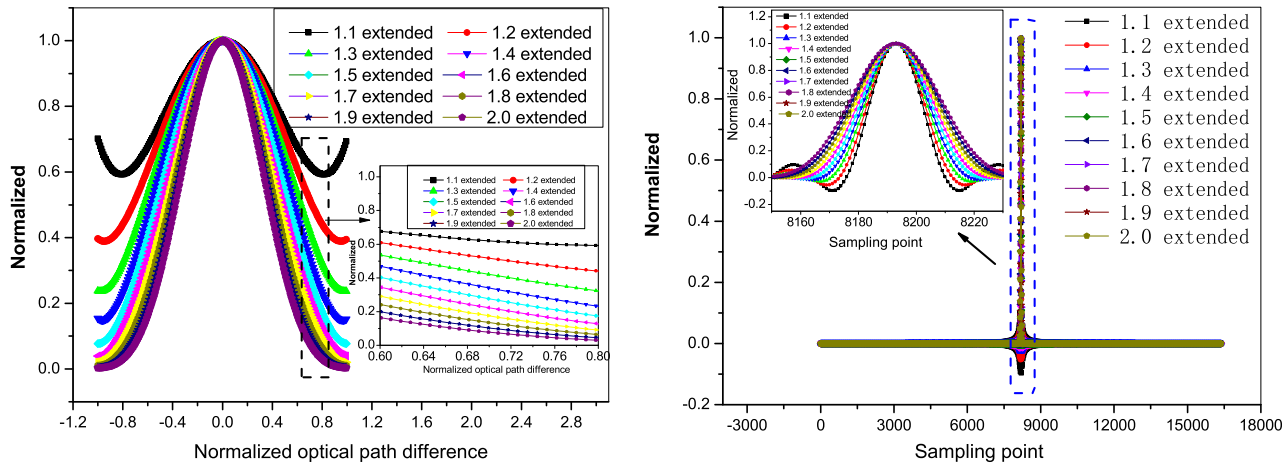


Fig. 12. Apodization function. (Left) Optimized apodizing functions with different extents. (Right) Spectra recovered by optimized apodizing functions.

than that without the apodization function, and the interval is 0.1, as shown in Fig. 12 (right). The restored spectra corresponding to ten kinds of apodization functions are shown in Fig. 12. From the enlarged image in the central region of the spectrum, it can be seen that the degree of sidelobe oscillation decreases with increasing apodization degree, and the FWHM of the spectrum increases gradually. In the GMI ground data processing system, the Level 1 spectral data processing process does not use apodization function by default. Users of different Level 1 data products can obtain Level 1 spectral data products with different apodization degrees by ordering according to application requirements.

C. Spectrum Recovery

Atmospheric GHG inversion is based on the spectrum. It is necessary to transform the interferogram data into the spectral domain to obtain the effective spectral information after recovery. Then, the digital value (DN) of spectral data is converted into the radiance value with physical significance through the application of spectral and radiometric calibration coefficients. The spectral restoration algorithm is designed as follows.

1) The ZPD of the interference fringe is accurately located, and the conjugate symmetry of the interference data is processed with the ZPD as the center and the long half-edge interferogram as the data.
2) According to the application requirement, the zero-charge value of the interference fringe is input, and the interferogram is extended to improve the details of the restored spectrum.
3) The FFT of the interferogram is used to obtain digital spectral data.
4) The spectral calibration coefficients are interpolated to match the data points of the spectral calibration coefficients and spectral calibration coefficients. After the data points are matched, the abscissa ordinates of the spectral data are replaced by the spectral calibration coefficients to complete the spectral calibration.
5) The radiation calibration coefficients are also interpolated to match the spectral data with the data points of the radiation calibration coefficients. The calibration coefficients (offset and response) after data matching are calibrated, and the calibrated spectra are output to form the Level 1 product data.

IV. Geometric Correction Algorithms

The geometry information of the observation points is obtained by theoretical calculation based on satellite position, scanning mirror angle, and relevant coordinate system. Due to the uncertain parameters of the satellite platform and GMI, there may be some deviation in geometry information. Because the original data obtained by the GMI are interferograms, the traditional geometric correction method based on GCPs cannot be used.

In this article, according to the response characteristics of the GMI interferograms for different surfaces, a coastline observation point is selected as the correction reference. The coastline has a large uniform surface and the albedo has a sudden change from land to sea, which is reflected in the interferogram, in which the average DN value over the sea is much smaller than that over the land. If the observed DN value is between the two, the observation point is likely to be located on the coastline, and the features of the interferogram obtained by the GMI in this area can be used for geometric correction.

A. Methodology

For GMI geometric correction, a satellite image with high geometric accuracy should be selected as the base map, and the geometric information of the coastal area pixels should be taken as the real latitude and longitude values. Therefore, an image from the wide field imager (WFI) on the CBERS-04 satellite is selected as the base map for geometric correction, as this image has a spatial resolution of 73 m and a geometric correction accuracy of 2 m [13]. Because the GMI observation point is projected on the Earth's surface in a circle

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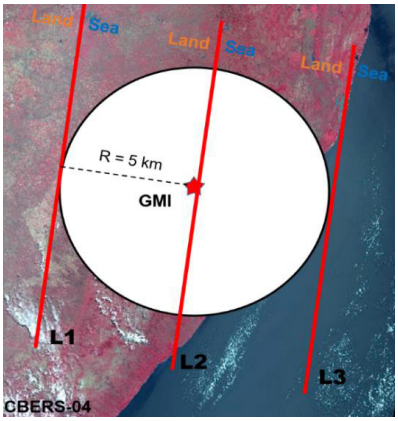


Fig. 13. GMI observation point on coastline area.

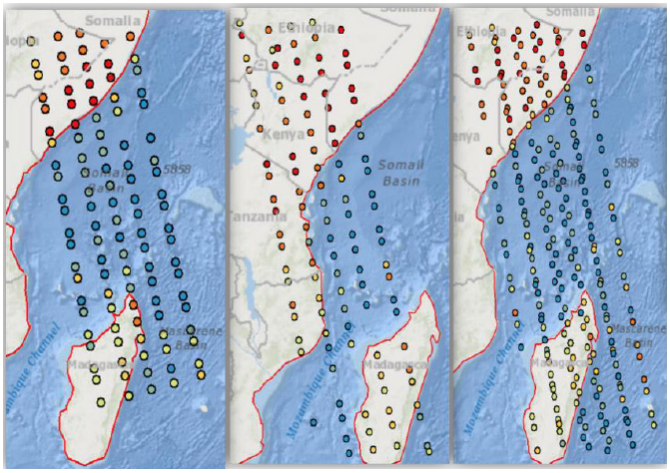


Fig. 14. Distribution of GMI observation points on the East coast of Africa (September 2–4, 2018).

with a radius of approximately 5 km, the geometric error of the base map is negligible.

In the detection of the observation points near the coastline area, the intersection of the GMI observation point and the coastline can be divided into three cases, corresponding to L1 (the observation point is tangent to the coastline and is located on the land), L2 (the observation point is tangent to the coastline), and L3 (the observation point is tangent to the coastline and is located in the ocean) in Fig. 13. In the NIR band, the reflectivity of the ocean is weak (L3 position), the reflectivity of the land is high (L1 position), and the reflectivity of the junction of the ocean and the land changes from a large area to a small area (L3 position). Therefore, when passing through L2 from L1 to L3, the DN of the interferogram changes greatly. It used the feature of large area uniform landform and abrupt change in the surface reflectance in coastline area of the high spatial resolution remote sensing image. This method realized the matching application between noncoaxial imaging remote sensing equipment and nonimaging remote sensing equipment. Only the observation point in the L2 case can be used for geometric correction, so this type of observation point is detected first.

If the averaged DN value of the land interferogram is I_{L} and the ocean interferogram is I_{S}, the calculation formula of the coastline interferogram I_{C} is as follows:

$$I_c = a \cdot I_L + b \cdot I_S \tag{4}$$

$$a + b = 1. \tag{5}$$

Parameters a and b are the proportions of land and ocean in the field of view of the observation point, respectively, and are obtained according to the statistics of the GMI observation points. If the average DN value of the interferogram of the observation point is I_j, then the two points near this point in the same track direction are I_{j+1} and I_{j-1}, respectively, if I_j meets the following conditions:

$$I_{j-1} \le I_j \le I_{j+1}, \quad I_j \in [T_1, T_2] \tag{6}$$

or

$$I_{j-1} \ge I_j \ge I_{j+1}, \quad I_j \in [T_1, T_2] \tag{7}$$

where T_1 and T_2 are the thresholds of the averaged DN values of ocean and land interferograms, respectively, and it can be judged that point I_j is located on the coastline. To determine the threshold value in the judgment, it is necessary to calculate the statistics of the actual observed interferogram DN values on land and sea.

B. Correction Results

Fig. 14 shows the average DN value distribution of the GMI interferogram with the observation time from September 2 to 4, 2018. The colors of the observation points from blue to red represent an increase in the average DN value. After removing the abnormal data affected by clouds, the average DN value of most points on the ocean is less than 3000, while the average DN value of points on the land is more than 8000.

After the threshold values of land and sea interferograms are determined, the observation points on the coastline can be detected according to the above criteria. On the base map, the adjacent points to the left and right of the coastline observation point are connected, and the intersection point with the coastline is detected. The longitude and latitude of the intersection point are the real geometric information (lon′, lat′) and are subtracted from the theoretical longitude and latitude (lon, lat) to obtain the geometric error (δlon,δlat) of the GMI observation point. The coastline observation points are detected from the observation points in Fig. 14, and a total of 18 points are obtained. After manual identification, duplicate geographical position points were removed, and 12 coastline observation points were finally obtained. The geometric correction data of the coastline observation points are shown in Table II.

The least squares algorithm is used to fit the geometric correction data of each point linearly, and the GMI longitude and latitude geometric correction formulas are as follows:

$$\mathrm{lon}' = 1.00154 \cdot \mathrm{lon} + 0.00119 \tag{8}$$

$$\mathrm{lat}' = 0.98858 \cdot \mathrm{lat} + 0.01147. \tag{9}$$

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TABLE II
GEOMETRIC CORRECTION STATISTICS OF GMI COASTLINE OBSERVATION POINTS

Point	GMI		WFI		Error	
	lon	*lat*	*lon*	*lat*	δlon	δlat
1	-20.710	48.610	-20.709	48.481	-0.001	0.129
2	-0.707	42.315	-0.663	42.277	-0.044	0.038
3	0.611	43.777	0.664	43.595	-0.053	0.182
4	-3.126	40.362	-3.105	40.131	-0.021	0.231
5	-20.749	48.419	-20.735	48.471	-0.014	-0.052
6	-13.723	40.546	-13.716	40.588	-0.007	-0.042
7	-0.205	42.681	-0.207	42.685	0.002	-0.004
8	0.560	43.643	0.657	43.581	-0.097	0.062
9	-2.765	40.145	-2.778	40.224	0.013	-0.079
10	-5.714	38.873	-5.677	38.820	-0.037	0.053
11	-7.262	39.195	-7.218	39.404	-0.044	-0.209
12	-9.934	39.751	-9.874	39.861	-0.060	-0.110

V. LEVEL 1 PRODUCT RESULTS

TheGF-5 satellite travels in a sun-synchronous orbit at an altitude of 705 km with an equatorial crossing local time descending node (LTDN) of 13:30. This satellite completes an orbit in approximately 100 min and operates on a global basis with a 7-day orbit repeat cycle but with a 51-day footprint revisit cycle. The measurement mode of the GMI contains a grid point observation mode and a sun-glint mode. During grid point observations, the GMI uses a scanning mirror in a flexible scan pattern with 1–9 cross-track points within +/−35°of nadir, yielding an ∼750-km cross-track range, as shown in Fig. 15(a), which presents the characteristics of the observation patterns at about north at a latitude of 30°. The GMI has a field of view of 14.6 mrad, which corresponds to a surface footprint with a diameter of 10.5 km at nadir. After extensive on-orbit performance tests, the five-point cross-track scan mode was used from May 2018 to the present day for routine scientific observations. After the GMI is normally started and making observations, the ground processing system immediately produces the Level 1 products. Fig. 15(b) shows the ground track of the first flight observation over China.

An observation point located in the Yangtze River Delta area (longitude 121.550, latitude 31.234) is selected for processing. Based on the Level 1 processing algorithm, the quality control, error correction, and spectral recovery of the interferogram are processed, and the geometric information, spectral, and radiometric calibration coefficients are loaded. Finally, the Level 1 products (radiance spectrum) of GMI Band 1 to Band 4 are shown in Fig. 16, which is also the first hyperspectral GHG spectrum product obtained by the SHS technology on a satellite platform.

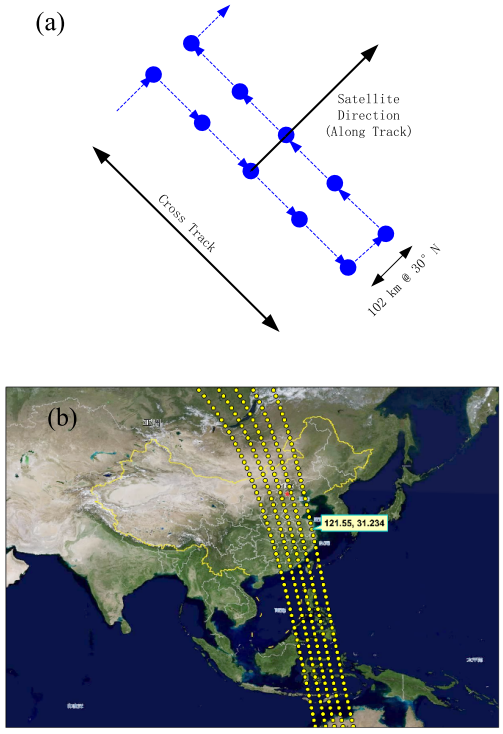


Fig. 15. GMI grid point observations in the five-point cross-track scan mode. (a) Grid observation (five points). (b) GMI ground track.

A. Spectral Resolution

The ILS function represents the response of the instrument to a monochromatic light source, and its FWHM value is the spectral resolution of the instrument. The ILS calibration of GOSAT and TANSO-FTS is performed with the onboard diode laser [14]. However, the GMI does not carry a laser. It is known that the solar spectrum obtained by radiometric calibration includes many Fraunhofer lines. It is not affected by atmospheric and surface factors but is only related to the convolution of the instrument. Therefore, solar Fraunhofer lines are selected as the GMI ILS calibration source, which needs to meet the independent spectral line and certain absorption depth. To improve the detection accuracy, two solar Fraunhofer lines are selected as calibration sources for each GMI band.

By changing the parameters of the ILS obtained from the ground test and iterating with the least square optimization,

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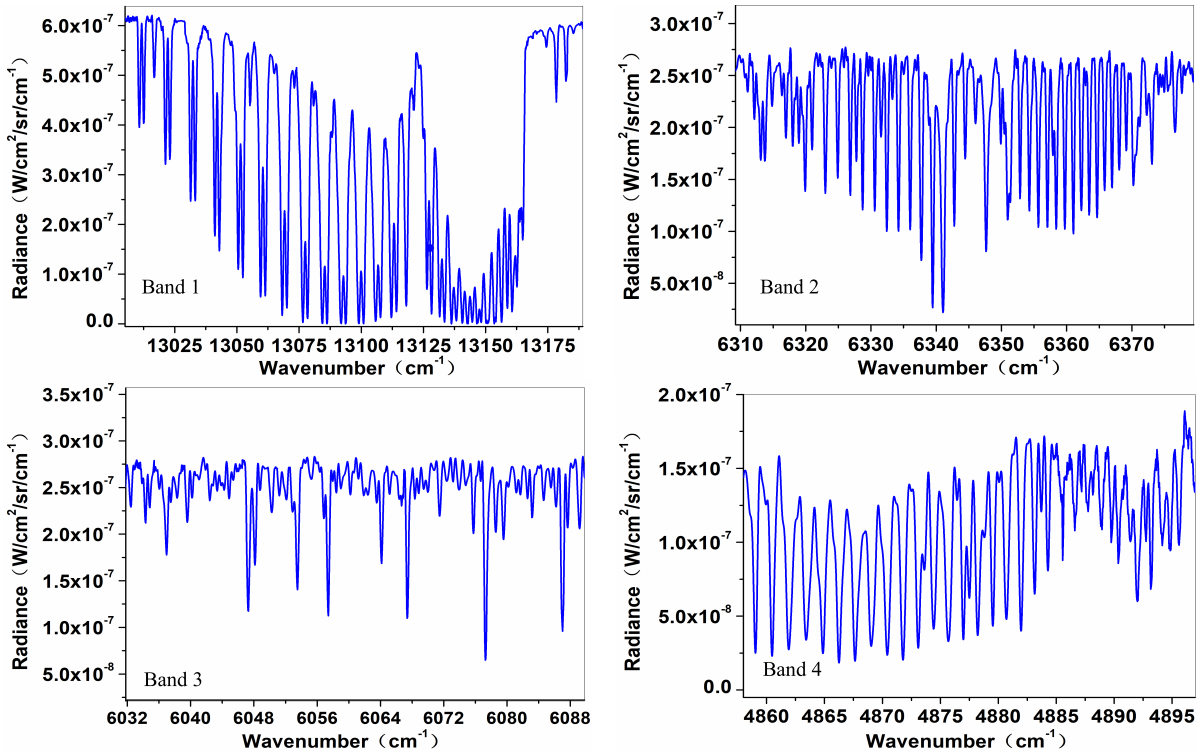


Fig. 16. GMI first results of the Level 1 products (longitude 121.550, latitude 31.234).

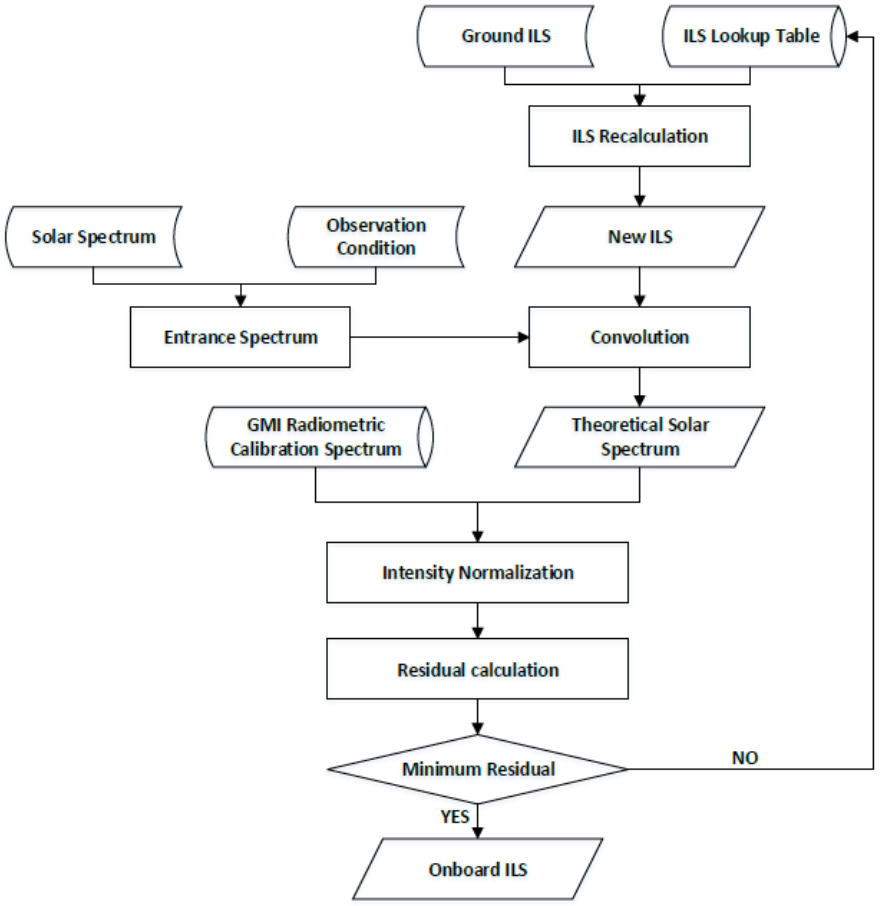


Fig. 17. GMI ILS calibration flowchart.

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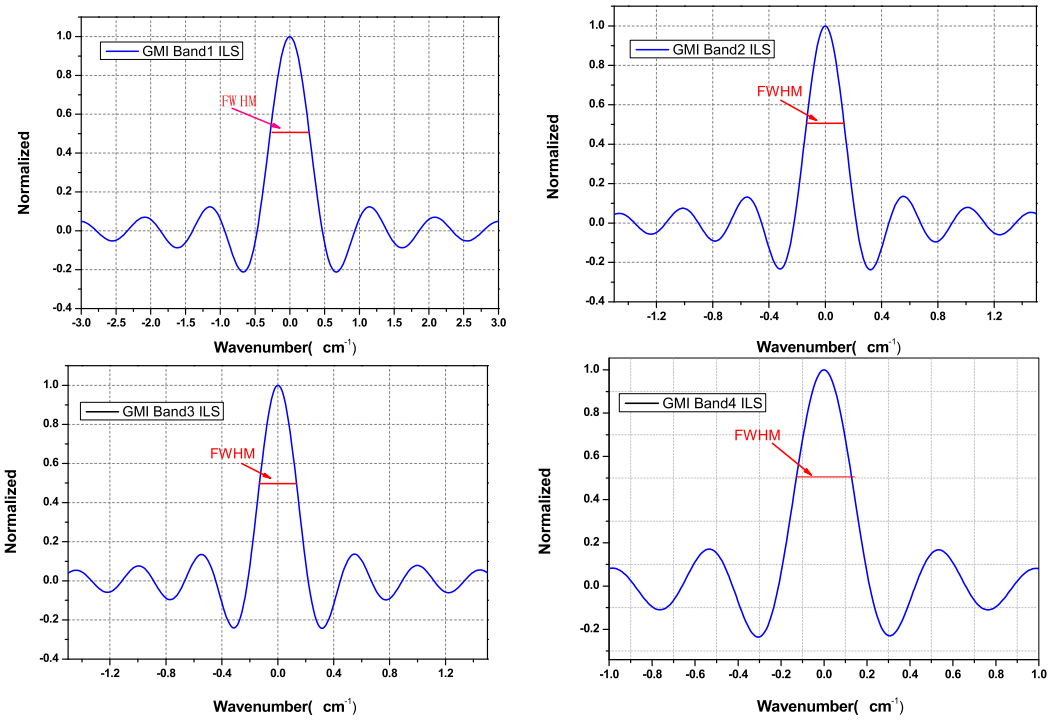


Fig. 18. GMI onboard ILS.

the actual ILS is achieved when the residual error between the simulation spectrum and the measured spectrum is the smallest. A detailed flowchart of the ILS calibration is shown in Fig. 17. Through iterative calculation, the results of the onboard ILS calibration of the GMI are obtained, as shown in Fig. 18. By measuring the FWHM value of each ILS band, the spectral resolutions of Band 1–Band 4 are 0.572, 0.262, 0.263, and 0.266 cm^{-1}, respectively, which are better than the requirements of the instrument.

The main possible change in ILS is FWHM. Comparing the FWHM results of the ILS before and after launch, it shows that the FWHM is basically unchanged. This is because the ILS calibration test before launch is carried out in simulated on-orbit environment, and the GMI on-orbit thermal control system works well, especially the temperature control of interferometer components.

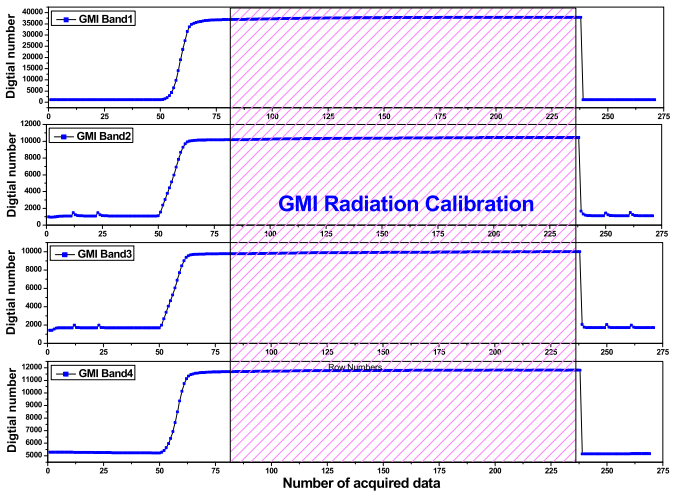


Fig. 19. GMI radiation calibration data statistics.

B. SNR

The SNR is a critical parameter to evaluate satellite observation. The SNR is a measurement that compares the level of a desired signal to the level of background noise. The SNR indicates how useful the data may be for a given measurement. The proper treatment of measurement noise is the dominant consideration in designing retrieval methods. In the laboratory characterization of the GMI, a 300-cm-diameter integrating sphere with a 100-cm-diameter exit port was used for SNR testing.

A stable and uniform radiation source is needed for the SNR test. Therefore, the data obtained in the GMI radiation calibration stage are selected for on-orbit SNR measurements. According to the Sun–Earth-satellite angle O_{SES}, the incident angle O_{DP} to the diffuser plate can be calculated. In the period of obtaining GMI radiometric calibration data (approximately 3–4 min), the change in the O_{DP} angle is approximately 0.2°, and the corresponding change in the radiation intensity is approximately 0.1%. To verify the theoretical estimation, the interferograms obtained in the GMI calibration stage are statistically analyzed, and the average value of each interferogram is taken as the response value of this measurement, as shown in Fig. 19. The statistical results show that the intensity changes in the radiometric calibration are consistent with the theoretical estimates.

The SNR of the GMI represents the SNR of the spectrum, not for the interferogram. First, it is necessary to process the

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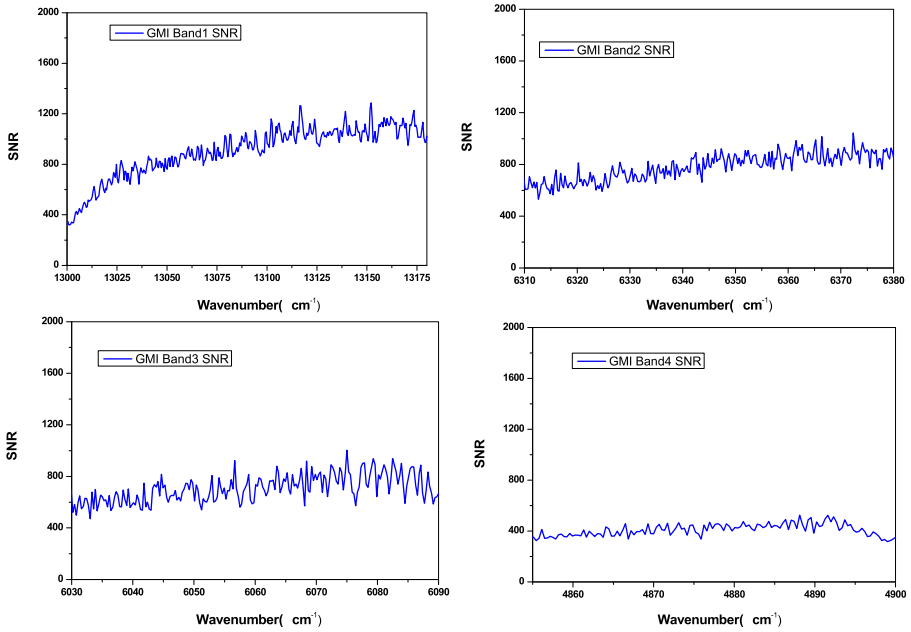


Fig. 20. GMI onboard SNR.

test data with the Level 1 product. The calculation of the SNR is very similar to that performed for most optical remote sensing instruments. The SNR was characterized using the following equation at each energy level of the radiance for each channel of the four bands:

$$\mathrm{SNR}_{\mathrm{cal}} = \frac{S_{\mathrm{mean}}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(S_i - S_{\mathrm{mean}})^2}} \tag{10}$$

where $\mathrm{SNR}_{\mathrm{cal}}$ is the signal-to-noise ratio of one channel at a given energy level, S_{mean} is the average radiant intensity from n samplings at a given energy level, n is the number of samplings of radiant intensity at a given energy level, and S_i is one sampling of radiant intensity at a given energy level. The energy level corresponding to $\mathrm{SNR}_{\mathrm{cal}}$is inconsistent with the requirements in Table I. For the $\mathrm{SNR}_{\mathrm{orb}}$ condition, the conversion calculation of SNR is required. The formula is as follows:

$$\mathrm{SNR}_{\mathrm{orb}} = \mathrm{SNR}_{\mathrm{cal}}\sqrt{\frac{L_{\mathrm{orb}}}{L_{\mathrm{cal}}}}. \tag{11}$$

$L_{\mathrm{cal}}$is the energy level of the GMI radiation calibration, and L_{orb} is the reflected energy level of the Earth's surface at an albedo of 0.3 and sun elevation angle of 30°. Fig. 20 illustrates the results of the onboard-measured SNR for each band, which meet the mission requirements. The channel average SNR values for each band at the required intensity are approximately 910 for Band 1, 680 for Band 2, 670 for Band 3, and 360 for Band 4.

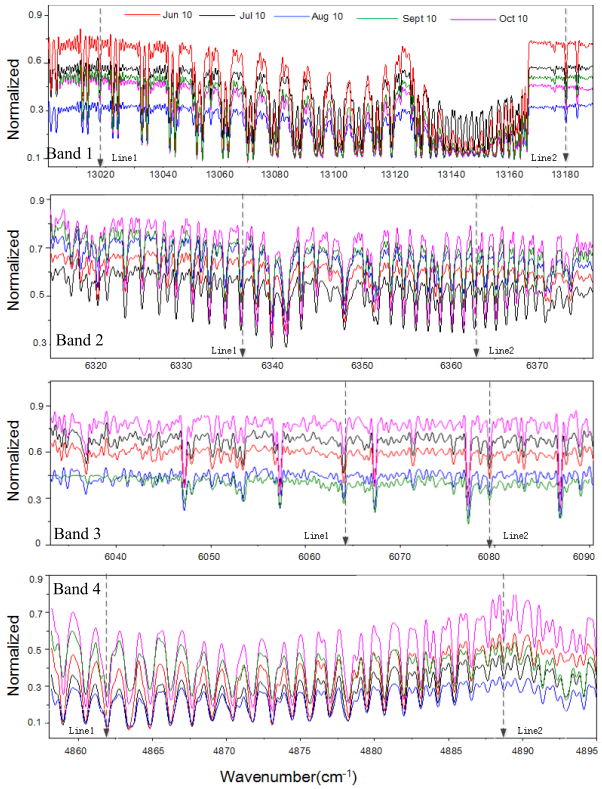


Fig. 21. GMI spectral stability.

C. Spectral Stability

To acquire accuracy GHGs' retrieval, in addition to the need for high spectral resolution and a high SNR spectrum, the GMI also has to have stable Level 1 product performance, especially spectral and radiation characteristics. To test the stability of GMI-observed spectrum, the wavelength shifts of five-month GMI-observed spectrum are evaluated by comparing with two independent molecular absorption lines (Line 1 and Line 2) in all four bands. Fig. 21 shows the five sampled observations and the central position of Line 1 and Line 2 in every band.

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TABLE III
LEVEL 1 PRODUCT WAVENUMBER VALUE DRIFT STATISTICS

Time (2018)	Band 1 (cm^{-1})		Band 2 (cm^{-1})		Band 3 (cm^{-1})		Band 4 (cm^{-1})	
	Line 1	Line 2	Line 1	Line 2	Line 1	Line 2	Line 1	Line 2
Jun 10	13017.9790	13179.9970	6336.4353	6334.6466	6064.0515	6079.4725	4861.9178	4887.6720
Jul 10	13017.9900	13180.0270	6336.4363	6334.6536	6064.0415	6079.4736	4861.9245	4887.6520
Aug 10	13017.9900	13179.9970	6336.4363	6334.6536	6064.0515	6079.4746	4861.9245	4887.6520
Sept 10	13017.9900	13180.0270	6336.4473	6334.6436	6064.0445	6079.4857	4861.9245	4887.6554
Oct 10	13017.9900	13179.9970	6336.4363	6334.6536	6064.0515	6079.4857	4861.9111	4887.6654
Maximum deviation	0.0200	0.0300	0.0110	0.0102	0.0070	0.0130	0.0130	0.0200

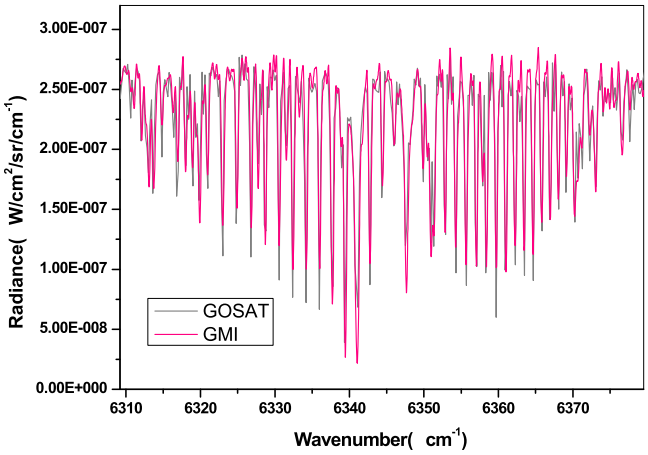


Fig. 22. Comparison of GMI and GOSAT spectra.

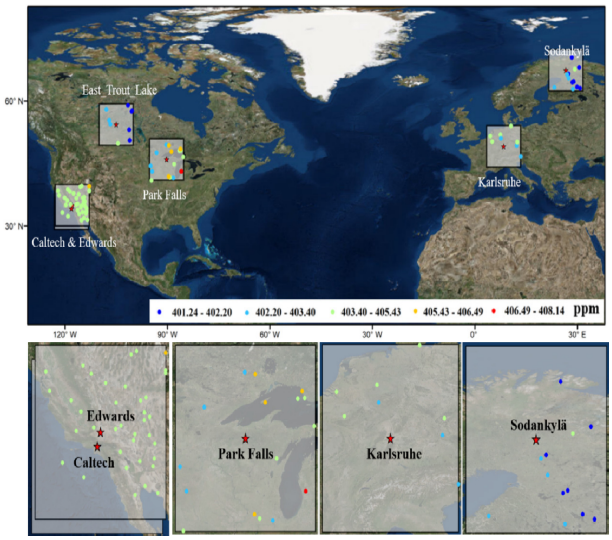


Fig. 23. TCCON verification stations and GMI observation points.

Theoretically, the wavelength position of the molecular spectrum remains constant, so it can be used to evaluate the spectral stability of the instrument. Comparing with the selected molecular absorption lines, the maximum wavelength shifts are calculated in all bands. Table III lists the detail. The result shows that the wavelength shifts are 0.03, 0.011, 0.013, and 0.02 cm^{-1}, respectively, from Band 1–Band 4.

Select GOSAT and GMI observation data near the same area for comparison, as shown in Fig. 22. The independent absorption lines are consistent, and the difference in radiation brightness is small. However, the spectral resolution of the two instruments is not the same, so there is a certain difference in the depth of the absorption peak.

VI. RETRIEVAL RESULTS

The TCCON is a network of ground-based Fourier transform spectrometers. Its measurements of GHGs are usually used to validate the same product from kinds of satellite missions, for example, AIRS, SCIAMACHY, IASI, GOSAT, and OCO-2. The precision of the resulting mole fractions retrieved from single spectra is approximately 0.15% for CO_2 and 0.2% for CH_4 [15]. To verify the retrieval capability of the GMI Level 1 products, some GMI observation points near the TCCON stations are selected to carry out retrieval experiments. The data selected meet the requirements that the difference between longitude and latitude is within 5° and that the time difference between observations is not more than 2 h. According to the operation and global distribution of TCCON stations, five verification stations are selected, as shown in Fig. 23. The red stars represent the TCCON stations, and the dots represent the GMI observation points near the stations.

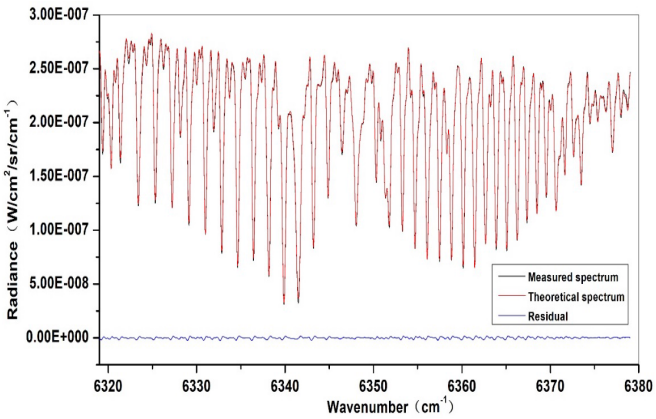


Fig. 24. Comparison between the theoretical spectrum and the measured spectrum (Band 2).

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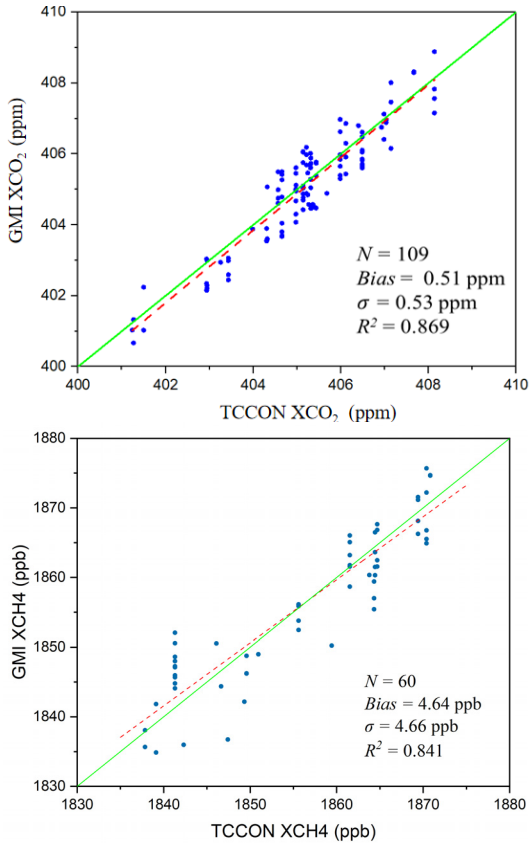


Fig. 25. Comparison of the initial GMI retrieval results with the TCCON data.

The main process of GMI retrieval experiment includes theoretical spectral calculation, accurate correction of atmospheric absorption lines, inversion method, and so on and finally obtains the retrieval results of XCO_2 (the column-averaged dry air mole fraction of CO_2) and XCH_4 (the column-averaged dry air mole fraction of CH4). To verify the quality of the Level 1 products, the theoretical spectrum of the observation point is simulated using the radiation transfer software SCIATRAN. Fig. 24 shows the superposition of the GMI Band 2-measured spectrum and theoretical spectrum, and the rms value of the two spectral residuals is approximately 0.8%, which indicates good agreement.

In this article, the inversion method is based on the OEM [16]–[18], which constructs an optimal solution x of the inverse problem by minimizing the *a posteriori* cost function and has been widely applied to the inversion of atmospheric parameters

$$\boldsymbol{x}_{i+1} = \boldsymbol{x}_a + \left(\boldsymbol{K}_i^{\mathrm{T}} \boldsymbol{S}_\varepsilon^{-1} \boldsymbol{K}_i + \boldsymbol{S}_a^{-1}\right)^{-1} \boldsymbol{K}_i^{\mathrm{T}} \boldsymbol{S}_\varepsilon^{-1} \times \left[\boldsymbol{y} - F(\boldsymbol{x}_i) - \boldsymbol{K}_i(\boldsymbol{x}_i - \boldsymbol{x}_a)\right] \quad (12)$$

where $\boldsymbol{x}$ represents the state vector XCO_2, subscript i represents the result of the ith iteration, $\boldsymbol{x}_a$ represents the initial guess value of the state vector, k represents the weight function, $\boldsymbol{s}_{\boldsymbol{\varepsilon}}$ represents the measurement error covariance matrix, $\boldsymbol{s}_a$ represents the prior covariance, y represents the measurement data, and F($\boldsymbol{x}_i$) represents the forward model data.

The 109 XCO_2 and 60 XCH_4 results are obtained on the retrieval experiment, as shown in Fig. 25. The linear correlation and the difference between the GMI and TCCON results are analyzed, and the goodness of fit (R^2), the mean of absolute error (bias), and the standard deviation of error (σ) are calculated. The results show that R^2 is 0.869, bias is 0.53 ppm, and σ is 0.53 ppm for XCO_2. Similarly, the results also show that R^2 is 0.841, bias is 4.64 ppb, and σ is 4.66 ppb for XCH_4. The initial retrieval results show that GMI Level 1 data products have excellent retrieval capability, and the column concentration information has good consistency with the TCCON station data

$$\text{Bias} = \frac{\sum_{j=1}^{N} (x_j - X_j)}{N} \quad (13)$$

$$\sigma = \sqrt{\frac{\sum_{j=1}^{N} (x_j - X_j - \text{Bias})}{N}} \quad (14)$$

where x is the GMI XCO_2 result, X is the TCCON result, j is the observation point, and N is the total number of the observation points.

VII. Conclusion

This study introduced a data processing algorithm for GMI, which is onboard the Chinese satellite GF-5. For the first time, the radiance spectrum was produced from the GMI observation interferogram, and the retrieved spectrum specifications were analyzed, including spectral resolution and SNR. According to the comparison of the measured and theoretical spectra, the results show that the RMS of the spectral residuals is approximately 0.8%, which indicates good consistency. In addition, the Level 1 products of the GMI within five months following observations have good spectral stability characteristics (less than 0.005 cm^{-1} for Band 1, 0.003 $cm^{-1}$for Band 2, 0.002 $cm^{-1}$for Band 3, and 0.004 $cm^{-1}$for Band 4), which lays the foundation for high-precision GHG inversion. Finally, based on the Level 1 products, a GHG retrieval experiment is carried out, and the results are compared with the TCCON station data. The initial results show that the standard deviation of error of XCO_2 is 0.53 ppm and that of XCH_4 is 4.66 ppb.

In the future work, GMI onboard observation data will be statistically analyzed, and the Level 1 product processing algorithms will be optimized, especially the real-time correction processing after the bad detector pixels are added. The retrieval algorithms will be improved, and more observation points will be retrieved. In addition, onboard calibration data will be processed to update the calibration coefficient and improve the data accuracy. Our team is also responsible for data processing of Level 2 products, and an extensive retrieval error characterization will be performed. Further validations will be performed through comparison with TCCON and other GHG satellite data involving large amounts of measured spectra collected during different periods and in different latitude bins.

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Observing Network (TCCON) data were obtained from the TCCON Data Archive hosted by the Carbon Dioxide Information Analysis Center, Lawrence Berkeley National Laboratory, USA (TCCON.onrl.gov).

REFERENCES

[1] Y. Z. Sun *et al.*, "GF-5 satellite: Overview and application prospects," *Spacecraft Recovery Remote Sens.*, vol. 39, no. 3, pp. 1–13, Jun. 2018.

[2] W. Xiong, "Greenhouse gases Monitoring Instrument (GMI) on GF-5 satellite (invited)," *Infr. Laser Eng.*, vol. 48, no. 3, pp. 24–30, Mar. 2019.

[3] A. Butz *et al.*, "Toward accurate CO_2 and CH_4 observations from GOSAT," *Geophys. Res. Lett.*, vol. 38, no. 14, pp. 1–6, Jul. 2011.

[4] D. M. Hammerling, A. M. Michalak, and S. R. Kawa, "Mapping of CO_2 at high spatiotemporal resolution using satellite observations: Global distributions from OCO-2," *J. Geophys. Res. Atmos.*, vol. 117, no. 6, pp. 1–10, Jun. 2012.

[5] J. M. Harlander, F. L. Roesler, J. G. Cardon, C. R. Englert, and R. R. Conway, "SHIMMER: A spatial heterodyne spectrometer for remote sensing of Earth's middle atmosphere," *Appl. Opt.*, vol. 41, no. 7, pp. 1343–1352, Jul. 2002.

[6] C. R. Englert *et al.*, "First results from the spatial heterodyne imager for mesospheric radicals (SHIMMER): Diurnal variation of mesospheric hydroxyl," *Geophys. Res. Lett.*, vol. 35, no. 19, pp. 116–122, 2008.

[7] C. R. Englert *et al.*, "Correction of phase distortion in spatial heterodyne spectroscopy," *Appl. Opt.*, vol. 43, no. 36, pp. 6680–6687, Dec. 2004.

[8] G. Hu *et al.*, "Spectral restoration method for spatial heterodyne Raman spectrometer," *J. Raman Spectrosc.*, vol. 48, no. 11, pp. 1633–1643, Nov. 2017.

[9] S. Ye, W. Xiong, Y.-L. Qiao, J. Hong, and Y.-H. Fang, "Data processing for interferogram of spatial heterodyne spectrometer," *J. Appl. Opt.*, vol. 29, no. 6, pp. 848–852, Jun. 2009.

[10] H. L. Shi, Z. W. Li, H. Y. Luo, and W. Xiong, "Non-uniformity of optical field after filter in spatial heterodyne spectrometer," *Acta Opt. Sinica*, vol. 36, no. 6, pp. 269–276, Jun. 2016.

[11] C. R. Englert and J. M. Harlander, "Flatfielding in spatial heterodyne spectroscopy," *Appl. Opt.*, vol. 45, no. 19, pp. 4583–4590, Sep. 2006.

[12] D. A. Naylor and M. K. Tahic, "Apodizing functions for Fourier transform spectroscopy," *J. Opt. Soc. Amer. A, Opt. Image Sci.*, vol. 24, no. 11, p. 3644, Nov. 2007.

[13] H. Wu, "Influence of cloud and aerosol in atmospheric CO_2 inversion and its correction method," Ph.D. dissertation, Chin. Acad. Sci., Hefei, China, 2019, pp. 88–89.

[14] A. Kuze, H. Suto, M. Nakajima, and T. Hamazaki, "Thermal and near infrared sensor for carbon observation Fourier-transform spectrometer on the Greenhouse Gases Observing Satellite for greenhouse gases monitoring," *Appl. Opt.*, vol. 48, no. 35, pp. 6716–6733, Dec. 2009.

[15] D. Wunch *et al.*, "Comparisons of the Orbiting Carbon Observatory-2 (OCO-2) XCO_2 measurements with TCCON," *Atmos. Meas. Techn.*, vol. 10, no. 6, pp. 1–45, Feb. 2016.

[16] C. D. Rodgers, "Retrieval of atmospheric temperature and composition from remote measurements of thermal radiation," *Rev. Geophys. Space Phys.*, vol. 14, no. 4, pp. 609–624, Jul. 1976.

[17] A. Doicu, F. Schreier, and M. Hess, "Iteratively regularized Gauss–Newton method for atmospheric remote sensing," *Comput. Phys. Commun.*, vol. 148, no. 2, pp. 214–226, Oct. 2002.

[18] F.-M. Bréon and P. Ciais, "Spaceborne remote sensing of greenhouse gas concentrations," *Comp. Rendus Geosci.*, vol. 342, nos. 4–5, pp. 412–424, Apr. 2010.



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